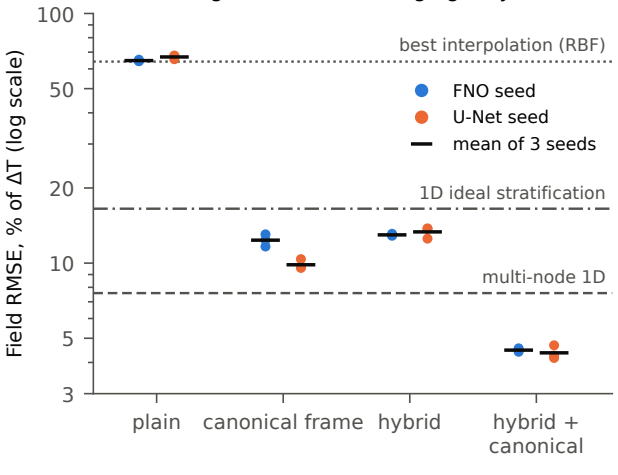
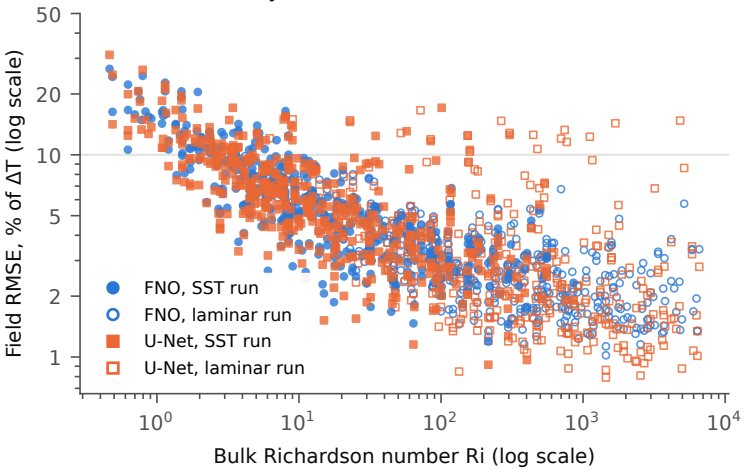

(a) Discharge, trained on charging only
Field RMSE, % of ΔT (log scale)
100
50
20
10
5
3
best interpolation (RBF)
1D ideal stratification
multi-node 1D
FNO seed
U-Net seed
mean of 3 seeds
plain
canonical frame
hybrid
hybrid + canonical


(b) Per test run, hybrid + canonical models
Field RMSE, % of ΔT (log scale)
50
20
10
5
2
1
FNO, SST run
FNO, laminar run
U-Net, SST run
U-Net, laminar run
$10^0$
$10^1$
$10^2$
$10^3$
$10^4$
Bulk Richardson number Ri (log scale)